

SCREENSHOTS OF ML FUNDAMENTAL VISUALISATIONS

1.DATASET SAVED:

```
*** student_performance.csv created successfully
```

	Study_Hours	Attendance	Assignments	Marks
0	1	60	2	35
1	2	65	3	40
2	3	70	4	45
3	4	72	5	50
4	5	75	5	58
5	6	80	6	65
6	7	82	7	72
7	8	85	7	78
8	9	88	8	82
9	10	90	8	88
10	11	92	9	92
11	12	95	10	96

2.DATASET EXPLORATION :

```
*** First 5 Rows
```

	Study_Hours	Attendance	Assignments	Marks
0	1	60	2	35
1	2	65	3	40
2	3	70	4	45
3	4	72	5	50
4	5	75	5	58

```
Last 5 Rows
```

	Study_Hours	Attendance	Assignments	Marks
7	8	85	7	78
8	9	88	8	82
9	10	90	8	88
10	11	92	9	92
11	12	95	10	96

```
Shape: (12, 4)
```

```
Columns:
Index(['Study_Hours', 'Attendance', 'Assignments', 'Marks'], dtype='object')
```

```
Data Types:
Study_Hours    int64
Attendance     int64
Assignments    int64
Marks          int64
dtype: object
```

```
Summary Statistics:
```

	Study_Hours	Attendance	Assignments	Marks
count	12.000000	12.000000	12.000000	12.000000
mean	6.500000	79.500000	6.166667	66.750000
std	3.605551	11.172206	2.443296	21.080474
min	1.000000	60.000000	2.000000	35.000000
25%	3.750000	71.500000	4.750000	48.750000
50%	6.500000	81.000000	6.500000	68.500000
75%	9.250000	88.500000	8.000000	83.500000
max	12.000000	95.000000	10.000000	96.000000

3.FEATURES AND LABELS :

Features (X):

	Study_Hours	Attendance	Assignments
0	1	60	2
1	2	65	3
2	3	70	4
3	4	72	5
4	5	75	5
5	6	80	6
6	7	82	7
7	8	85	7
8	9	88	8
9	10	90	8
10	11	92	9
11	12	95	10

Label (y):

0	35
1	40
2	45
3	50
4	58
5	65
6	72
7	78
8	82
9	88
10	92
11	96

Name: Marks, dtype: int64

4. TRAIN-TEST SPLIT :

```
Total Samples : 12
Training (80%) : 9
Testing (20%) : 3
```

```
x_train shape: (9, 3)
x_test shape: (3, 3)
y_train shape: (9,)
y_test shape: (3,)
```

5. TRAIN LINEAR REGRESSION :

```
=====
PART 5 - TRAIN LINEAR REGRESSION
=====
```

✅ Model Trained!

▴ Coefficients (Slope):

Study_Hours : 3.4549

Attendance : 1.3449

Assignments : -2.4027

▴ Intercept: -48.1006

Answers:

1. Slope → Coefficients of each feature.
Shows how much Marks change per unit increase.
2. Intercept → Value of Marks when all features are 0.
3. Meaning → Higher Study_Hours, Attendance & Assignments
→ Higher predicted Marks.

6. PREDICTIONS :

	Study_Hours	Attendance	Assignments	Predicted_Marks
0	5	80	5	64.753676
1	8	90	8	81.359291
2	12	95	10	97.097928

7. ACTUAL VS PREDICTED :

```
=====
PART 7 - ACTUAL VS PREDICTED
=====
      Actual_Marks  Predicted_Marks  Difference
0              92      92.011029    0.011029
1              88      88.269051    0.269051
2              35      31.243984    3.756016

! Highest Prediction Error: 2
  Error: 3.76 marks
```

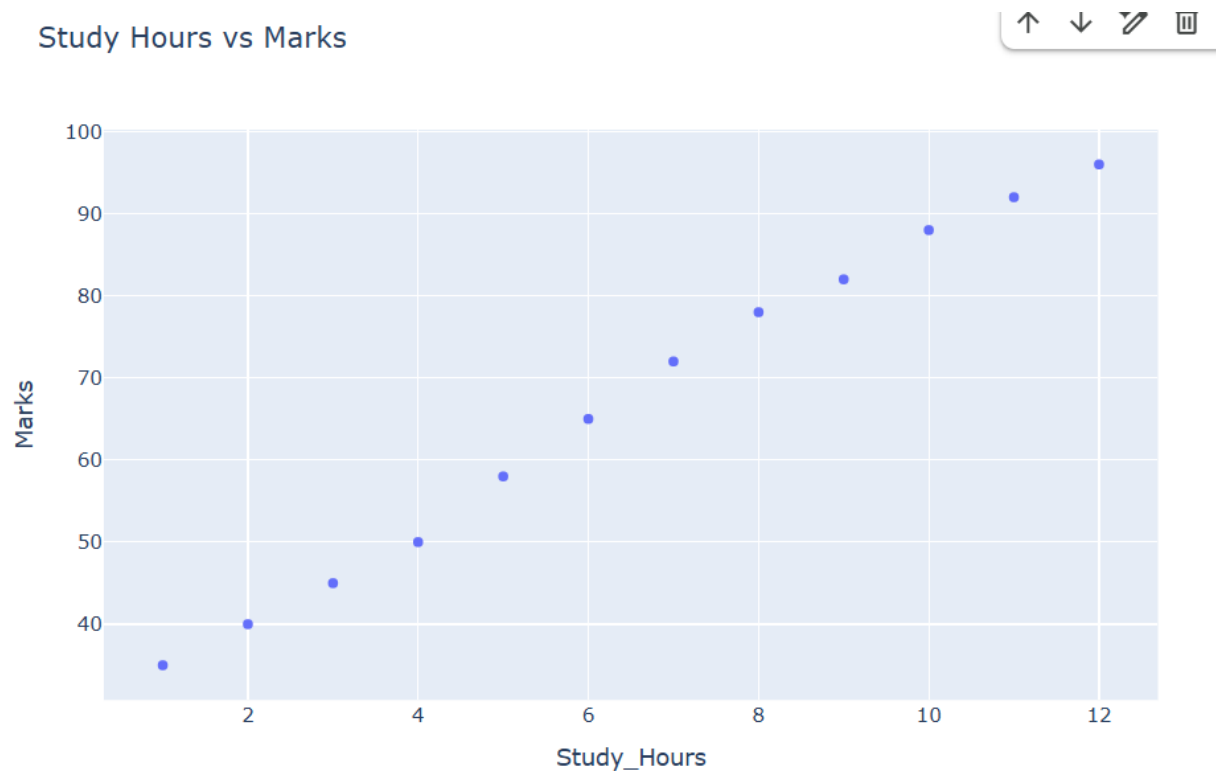
8. MODEL EVALUATION :

Mean Absolute Error (MAE): 1.3453654188948245

R² Score: 0.9929962957709836

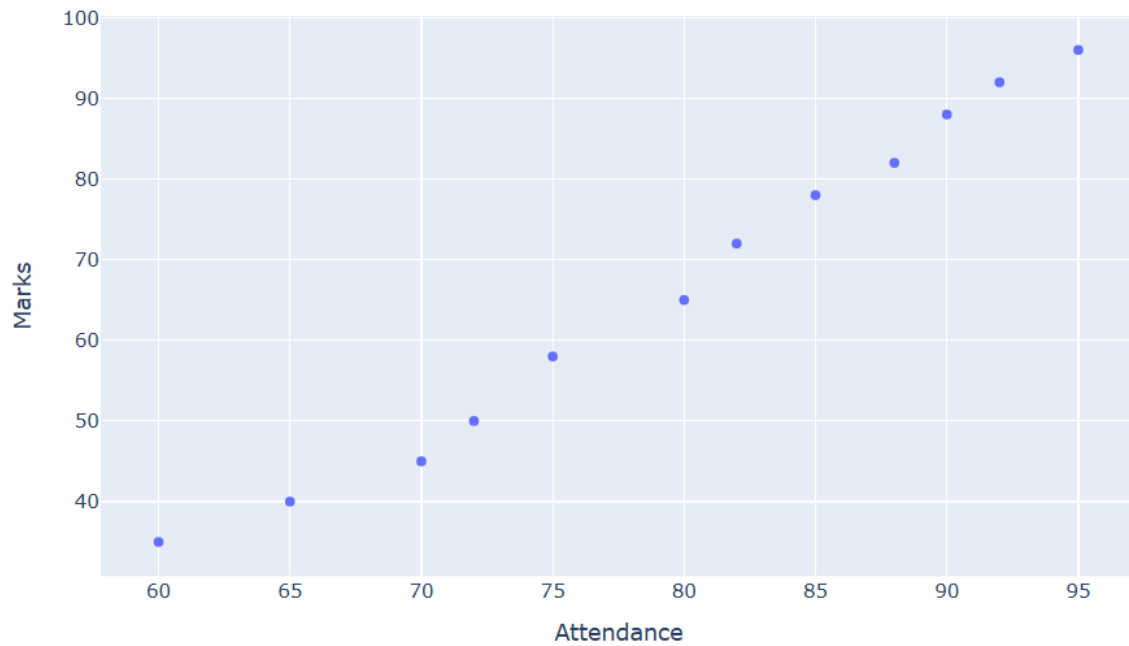
9. ISUALISATIONS :

9A. PLOT-1:STUDY HOURS VS MARKS :



9B.PLOT-2:ATTENDANCE VS MARKS :

Attendance vs Marks

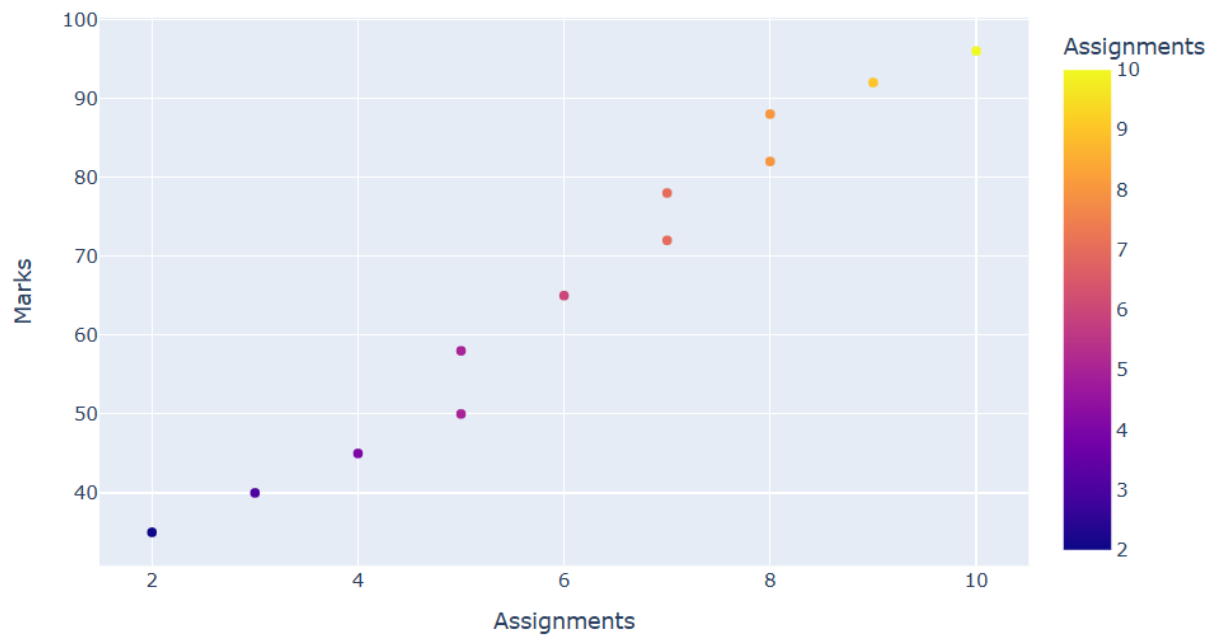


Terminal



9C . PLOT-3- ASSIGNMENTS VS MARKS :

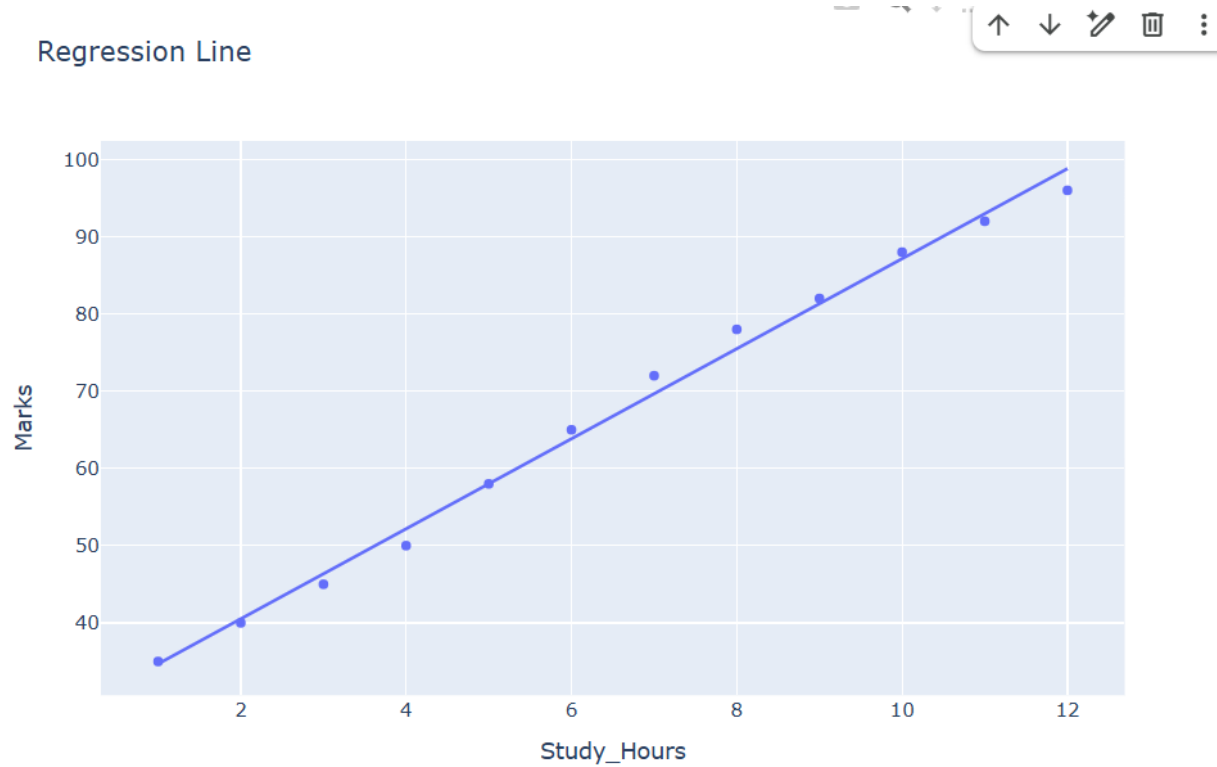
Assignments vs Marks



Terminal



9D.PLOT-4 -REGRESSION LINES :



9E .PLOT-5-PREDICTED MARKS TREND(STUDY HOURS 1-15)

Predicted Marks Trend

